

Summary

Software engineer and Yale PhD researcher building numerical, GPU-accelerated, and distributed applications in Python and C++. Experience spans CUDA/ROCm/HIP, JAX/PyTorch computational graphs, HPC systems, Linux development, and systematic gdb-based debugging.

Experience

Yale University, Electrical Engineering

PhD Research Assistant

2021–present

- Developed graph neural networks and automatic-differentiation models in Python to predict material properties including optical emission spectra.
- Developed Python and C++ software for Quantum ESPRESSO and post-DFT many-body simulation using MPI, OpenMP, PETSc, SLEPc, CUDA, ROCm/HIP, and CMake across leadership-class HPC systems.
- Sharded JAX/PyTorch computational-graph workloads across up to 800 Frontier GPUs and TB-scale datasets on national-lab GPU clusters including Frontier, Frontera, and Perlmutter.
- Used gdb for stack, frame, and memory inspection alongside cuda-gdb, nvprof, TotalView, and Linaro DDT for MPI/GPU distributed debugging and performance analysis.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS talks in 2024, 2025, and 2026, demonstrating research output alongside production-style software work.

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Built C++/CUDA parallel-computing applications with OpenCL fallback for large waveform and sensor-data processing, diagnostic analytics, and field-data visualization.
- Built C++ GUI applications with OpenGL custom plotting controls for rendering and interrogating large sensor datasets; used gdb to inspect stacks, frames, and memory during debugging.
- Implemented a supporting C#/ASP.NET web server and Microsoft SQL Server services for remote tool testing, calibration, data collection, and processing.

FindLight

Photonics Technical Writer

2018–2019

- Wrote technical content translating specialized optics and photonics topics into clear engineering-facing explanations.

Selected Projects

Machine Learning Models for Material Properties: Developed PyTorch Geometric graph neural networks and automatic-differentiation models for molecular-property prediction and optical-properties calculations in material systems.

Agentic Coding and Scientific Automation: Built LangChain-based workflows using OpenAI API models and local Ollama models to read documentation, generate structured inputs, and accelerate coding and research workflows.

Container Orchestration for First-Principles Computing: Developed CUDA-ready and cross-environment containers for first-principles software stacks and published them to Docker Hub for portable scientific computing.

Matrix-Multiply AI Accelerator: Developed a Verilog-based matrix-multiply accelerator project for AI-oriented compute pipelines and hardware-software performance exploration.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Technical Skills

Languages: Python, C, C++, CUDA, HIP, JAX, PyTorch, C#, Bash

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, NumPy, pandas, SciPy, SymPy, matplotlib

Systems: Linux, CMake, C++ GUI development, OpenGL custom plotting controls, Docker, Docker Compose, Kubernetes

GPU / Debug: CUDA, ROCm/HIP, cuBLAS, cuTENSOR, cuRAND, rocBLAS, rocFFT, gdb, cuda-gdb, nvprof, TotalView, Linaro DDT

Platforms: Frontier, Frontera, Perlmutter, Linux, CMake, Docker

ML / Automation: PyTorch, PyTorch Geometric, scikit-learn, LangChain, OpenAI API, Ollama, graph neural networks, transformers, agentic automation